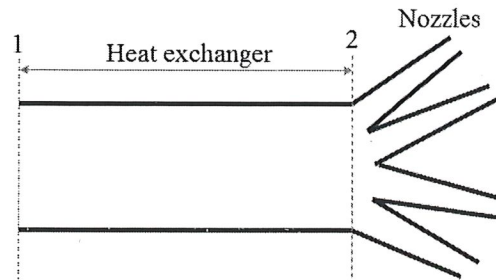


EXPLAINED EXAMPLE 4 – STEADY FLOW ENERGY EQUATION

①

An inventor is trying to develop a better industrial hair dryer for salons where a central heat exchanger heats the air and disperses it to a number of different drying stations. Her prototype for a small salon has four dryers with identical nozzles, as is shown in the figure below. Your job in the patent office is to analyze her device and determine whether or not this invention is a good idea to release to the public.



②

- a. The specifications on his patent application show that the working fluid should be air and the mass flow at the entrance to the heat exchanger is a steady 8 kg/s. Redraw a sketch of the device with a control volume, showing where the mass and energy flows are, and calculate the mass flow through the end of each nozzle.

①

- b. Air is sucked into the heat exchanger at 300 K and 0.2 MPa. 1 kg/s of water enters the heat exchanger as a subcooled liquid with a temperature of 500 K and a pressure of 10 MPa and leaves at the same pressure and a temperature of 380 K. Given these conditions, what is the temperature of the air leaving the heat exchanger? Make sure to list your assumptions.

Given: air

②

$$\begin{aligned}\dot{m}_a &= 8 \text{ kg/s} \\ T_1 &= 300 \text{ K} \\ P_1 &= 0.2 \text{ MPa}\end{aligned}$$

Water

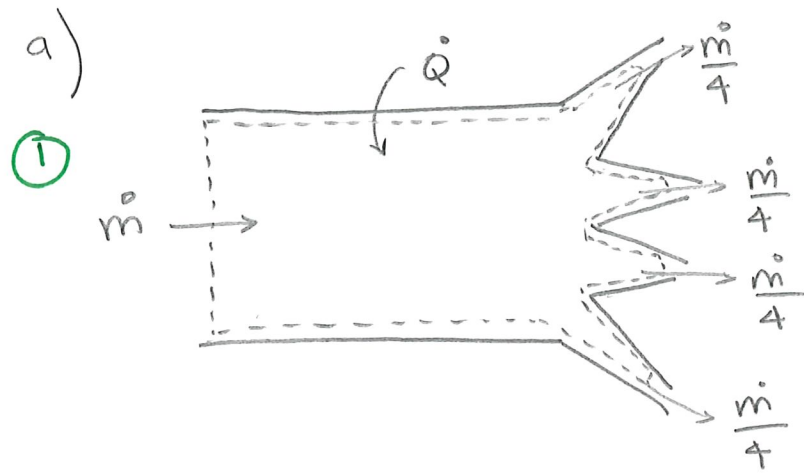
$$\begin{aligned}\dot{m} &= 1 \frac{\text{kg}}{\text{s}} \\ T_1 &= 500 \text{ K} \\ P_1 &= 10 \text{ MPa} \\ P_2 &= P_1 \\ T_2 &= 380 \text{ K}\end{aligned}$$

Assume: air is ideal

①

- : H_2O is S.C.S.
- : quasi-steady
- : $\Delta ke = \Delta pe = 0$
- : passive $\rightarrow \dot{W} = 0$
- : no heat lost to outside

Find: $T_{2, \text{air}}$



b) In air: $\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \dot{Q} - \dot{W}$

$\dot{m} \Delta h = \dot{Q}$

$\dot{m} c_p \Delta T = \dot{Q} \rightarrow T_2 = T_1 + \frac{\dot{Q}}{\dot{m} c_p}$ ← comes from water

In water: $\dot{m}(\Delta h + \cancel{\Delta ke} + \cancel{\Delta pe}) = \dot{Q} - \dot{W}$

$\dot{m} \Delta h = \dot{Q}$

→ State 1: $T_1 = 500 \text{ K}$
 $P_1 = 10 \text{ MPa}$ } subcooled liquid
 Table D.4B

$h_1 = 977.18 \frac{\text{kJ}}{\text{kg}}$

→ State 2: $T_2 = 380 \text{ K}$
 $P_2 = P_1 = 10 \text{ MPa}$ } Table D.4B

$h_2 = 455.37 \frac{\text{kJ}}{\text{kg}}$

② $\Rightarrow \dot{Q}_{\text{water}} = -\dot{Q}_{\text{air}} = \dot{m}(h_2 - h_1) = (1)(455.37 - 977.18)$
 $= -521.81 \frac{\text{kJ}}{\text{s}}$

$\Rightarrow \dot{Q}_{\text{air}} = 521.81 \frac{\text{kJ}}{\text{s}}$

① $\Rightarrow T_2 = T_1 + \frac{\dot{Q}}{\dot{m}C_p}$

$$= 300 + \frac{521810}{(8)(1001)} \rightarrow \boxed{T_2 = 365.16 \text{ K}}$$